

Mobile App Developer

Step 4: Work with APIs & Firebase

CareerCompass — Detailed Learning Notes

What this step is about

Connect mobile apps to remote data and backend services safely.

Core Concepts

- Understand HTTP methods, JSON, status codes, headers, authentication, timeouts, pagination and error handling.
- Create a service/data layer so UI code does not contain raw networking logic.
- Learn Firebase Authentication, Firestore/Realtime Database, Storage and Security Rules.
- Never place private secrets in a mobile client. Treat security rules and backend authorization as part of the design.

Skills You Should Be Able To Demonstrate

- Call and parse a REST API.
- Handle failed requests and slow networks.
- Implement authentication.
- Read/write user-specific Firestore data.
- Validate input before sending it.

Hands-on Project / Practice

- Build a Firebase-backed expense tracker with authentication, CRUD operations, categories and a summary screen.
- Add proper loading, empty and error states.

Recommended References

- Firebase Documentation — <https://firebase.google.com/docs>
- Firebase Full Course — <https://www.youtube.com/watch?v=fgdpvwEWJ9M>